📘 **AI/ML Internship – Day 15 Notes & Tasks**

🟢 **Python → AI/ML Thinking**

📌 **Part 1: What Changes from Today?**

👉 Till now:

- You learned Python syntax

👉 From now:

- You will work with **data**
- You will think like:
  - Data analyst
  - ML engineer

📌 **Part 2: What is Data?**

🔹 **Definition**

Data is a collection of information.

👉 Examples:

- Student marks
- Sales records
- Images
- Text

🔹 **Types of Data**

1. Numerical Data → numbers
2. Categorical Data → labels (Male/Female)
3. Text Data → sentences
4. Image Data → pixels

📌 Part 3: Data Representation in Python

🔹 Using List

```
marks = [80, 90, 70, 60]
```

🔹 Using Dictionary

```
student = {"name": "John", "marks": 80}
```

🔹 Using List of Dictionaries

```
students = [
    {"name": "A", "marks": 70},
    {"name": "B", "marks": 90}
]
```

👉 This is how real datasets look (basic form)

📌 Part 4: Basic Data Analysis (Manual)

🔹 Calculate Average

```
marks = [80, 90, 70]

avg = sum(marks) / len(marks)
print(avg)
```

🔹 Find Maximum

```
print(max(marks))
```

🔹 Count Values

```
count = len(marks)
print(count)
```

📌 Part 5: Data Cleaning (Important Concept)

🔹 **What is Data Cleaning?**

Removing incorrect or unwanted data.

🔹 **Example**

```
marks = [80, 90, None, 70]

cleaned = [m for m in marks if m is not None]
print(cleaned)
```

📌 **Part 6: Data Filtering**

```
marks = [80, 40, 90, 30]

passed = [m for m in marks if m >= 50]
print(passed)
```

📌 **Part 7: Data Transformation**

```
marks = [80, 90]

# Convert to percentage scale
scaled = [m / 100 for m in marks]
print(scaled)
```

📌 **Part 8: Why NumPy is Needed?**

👉 Problem now:

- We are using loops
- Code is slow for large data

👉 Solution:

- NumPy will:
    - Speed up processing
    - Handle large datasets

👉 So today builds foundation for tomorrow

📌 **Part 9: Real-World Thinking**

👉 In AI:

- Step 1 → Collect data
- Step 2 → Clean data
- Step 3 → Process data
- Step 4 → Train model

👉 Today = Step 1–3 (basic level)

🎯 **Tasks for Day 15**

🧠 **Task 1: Theory**

1. What is data?
2. Types of data
3. What is data cleaning?
4. What is data filtering?
5. Why NumPy is needed?

💻 **Task 2: Data Practice**

1. Create list of marks
2. Find average
3. Find max and min
4. Count total students

💻 **Task 3: Data Cleaning**

1. Remove None values
2. Remove negative values

3. Clean invalid data

💻 **Task 4: Data Filtering**

1. Find passed students
2. Filter marks > 75
3. Filter even numbers

💻 **Task 5: Data Transformation**

1. Convert marks to percentage
2. Multiply values
3. Normalize data